Dell Technologies | Cisco Distributor

# Licensing Deployment Scenarios

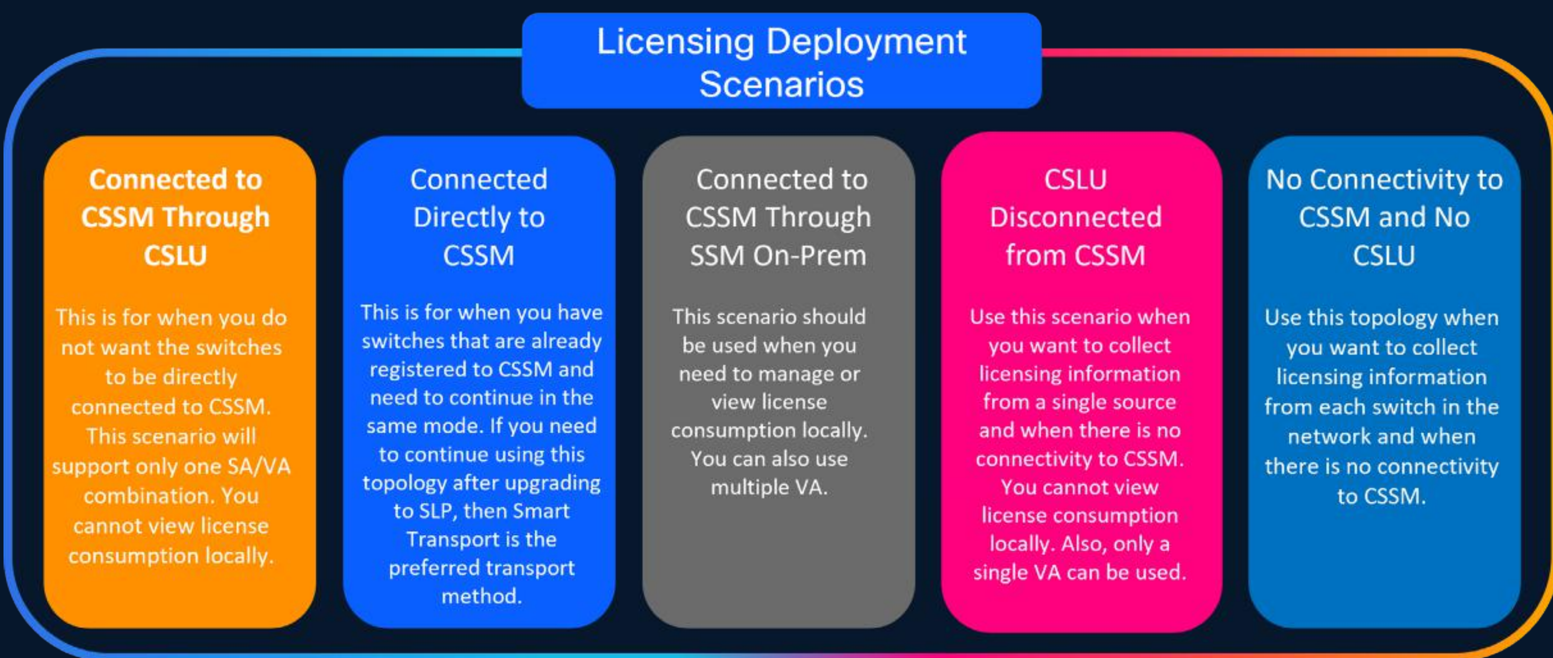
**This eFolio provides Dell engineers with a step-by-step reference for configuring Cisco Smart Licensing Using Policy (SLP) across five supported scenarios. From fully connected deployments to air-gapped environments, you'll find clear workflows, verification commands, and practical tips for selecting the right licensing model.**

**Select one of the tabs on the right or turn the page to get started!**

# Exploring Cisco Smart Licensing Scenarios

Smart Licensing Using Policy (SLP) supports multiple deployment scenarios to align with customer security requirements, connectivity restrictions, and operational needs. Understanding these models ensures Dell engineers can quickly determine the best licensing path, whether in fully connected environments or highly restricted, air-gapped networks.

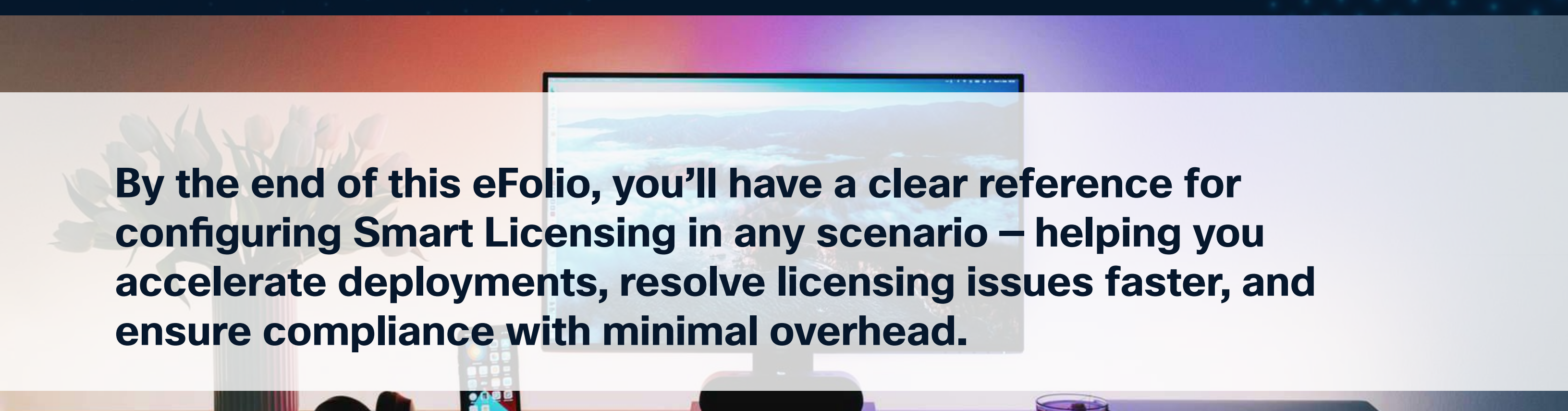
This guide expands on the five supported scenarios:



# What to Expect and Why These Scenarios Matter

Cisco Smart Licensing is a flexible, cloud-based software license model that simplifies how customers activate and manage their Cisco software. It replaces legacy licensing mechanisms that relied on Product Activation Keys (PAKs) and manual tracking.

| | |
|---|---|
| **Workflows** | ▪ Step-by-step configuration workflows – from switch commands to file transfers. |
| **Components** | ▪ Key components defined – acronyms are spelled out the first time they appear. |
| **Tips** | ▪ Verification tips – commands and outputs to confirm successful setup. |
| **Guidance** | ▪ Operational guidance – when to choose each topology based on customer environment. |



**By the end of this eFolio, you'll have a clear reference for configuring Smart Licensing in any scenario – helping you accelerate deployments, resolve licensing issues faster, and ensure compliance with minimal overhead.**

**Click here to view our Smart Licensing Guide**

# 1 Connected to Cisco Smart Software Manager (CSSM) Through Cisco Smart Licensing Utility (CSLU)

Here, switches in the network are connected to CSLU, and CSLU becomes the single point of interface with CSSM. A switch can be configured to push the required information to CSLU. Switch-initiated communication (push): A switch initiates communication with CSLU by connecting to a REST endpoint in CSLU. Data that is sent is unsecure and includes RUM reports.

Configure the switch to automatically send RUM reports to CSLU at required intervals. CSLU is the default method for a switch.

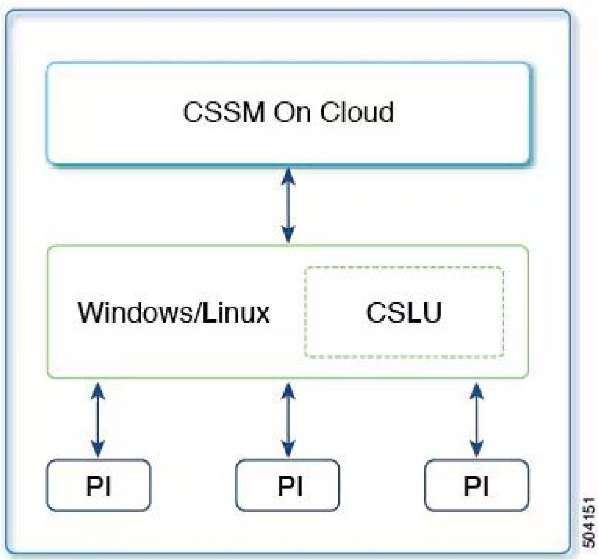



Figure 1Scenario: Connected to CSSM Through CSLU

## SLP Configuration – Connected to CSSM Through CSLU

| | |
|---|---|
| **Step 1: Install CSLU** | • Where the task is performed: .exe or .rpm file that you would download and deploy it as a VM as per your orchestration environment.<br>• Download the file from Smart Software Manager > Smart Licensing Utility.<br>• Refer to the Cisco Smart License Guide for help with installation and setup. |

## Step 2: Configure CSLU Preferences (in CSLU interface)

- Log in with Cisco ID.
- Add Smart Account (SA) and Virtual Account (VA)
- Add product instance (switch):
  1. Go to [https://software.cisco.com/download/home/286285506/type/286327971/release/](https://software.cisco.com/download/home/286285506/type/286327971/release/)
  2. Click the appropriate release.
  3. Under the Related Links and Documentation section, click **User Guide.**

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## Step 3: Configure Switch

Ensure network reachability between switch and CSLU. Set transport to:

switch(config)# license smart transport cslu

```
switch(config)# exit
switch# copy running-config startup-config
```

**Discover CSLU:**

- Option 1 (DNS zero-touch): No action if DNS resolves cslu-local
- Option 2 (DNS + domain): Same as above but with domain suffix
- Option 3 (Manual URL):

```
switch(config)# license smart url cslu http://<CSLU_IP>:8182/cslu/v1/pi
```

---

## Step 4: Verify

- Run:
  switch# show license status
- Check Next report push to confirm RUM report schedule.

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## Step 5: Flow

Switch → CSLU (RUM) → CSSM

CSSM → CSLU (ACK) → Switch

Click the information icon to view screenshots and examples.

**Click here to view our Smart Licensing Guide**

## 2 Connected Directly to Cisco Smart Software Manager (CSSM)

This method was available in the earlier version of Smart Licensing and remains supported with SLP.

Here, establish a direct and trusted connection from a switch to CSSM. The direct connection requires network reachability to CSSM. For the switch to then exchange messages and communicate with CSSM, configure one of the transport options available with this topology. Lastly, the establishment of trust requires the generation of a token from the corresponding Smart Account and Virtual Account in CSSM and installation on the switch.

You can configure a switch to communicate with CSSM in the following ways:

1. Use smart transport to communicate with CSSM (recommended)
2. Use Call Home to communicate with CSSM.

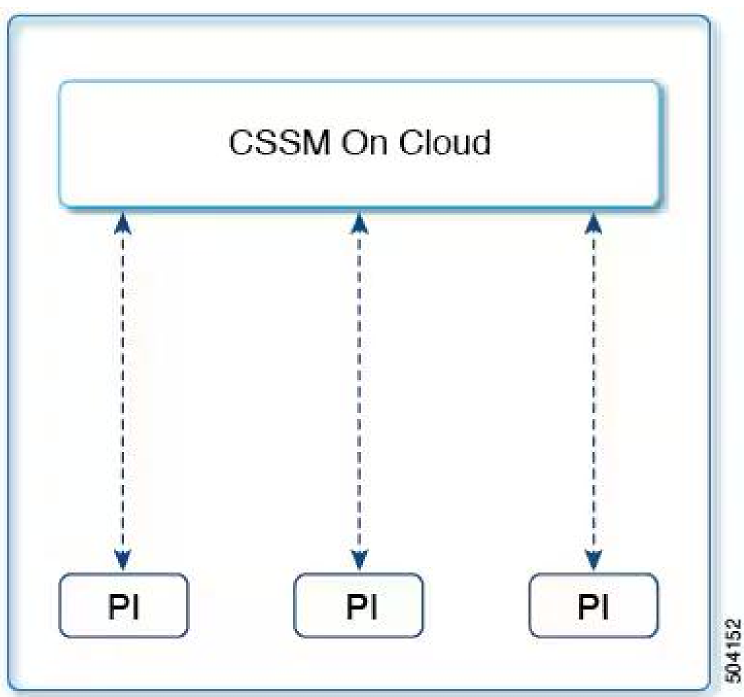

Figure 2 Scenario: Connected Directly to CSSM

### Smart Transport

Smart transport is a transport method where a Smart Licensing (JSON) message is contained within an HTTPs message and exchanged between a switch and CSSM to communicate.

- Smart transport: In this method, a switch uses a specific smart transport licensing server URL. This must be configured exactly as shown in the workflow section.
- Smart transport through an HTTPs proxy: In this method, a switch uses a proxy server to communicate with the CSSM.

### Call Home

Call Home provides email-based and web-based notification of critical system events. This method of connecting to CSSM was available in the earlier Smart Licensing environment and remains available with SLP.

- Direct cloud access: In this method, a switch sends usage information directly over the Internet to CSSM; no additional components are needed for the connection.
- Cloud access through an HTTPs proxy: In this method, a switch sends usage information over the Internet through a proxy server – either a Call Home Transport Gateway or an off-the-shelf proxy (such as Apache) to CSSM.

## SLP Configuration – Connected Directly to CSSM

### Step 1: Configure the Switch

### Choose one transport method:

1.1 Smart Transport (recommended):

switch(config)# license smart transport smart

switch(config)# license smart url smart https://smartreceiver.cisco.com/licservice/license

switch# copy running-config startup-config

1.2 Smart Transport via HTTPS Proxy:

switch(config)# license smart proxy <proxy_address> <port>

2.1 Call Home method: access.

Configure DNS and Call Home for either direct Internet or proxy gateway access. **Note:** Make sure that Smart Call Home is enabled on the switch before configuring Smart Software Licensing.

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### Step 2: Establish Trust with CSSM

In the CSSM Web UI:

Navigate to Inventory > Virtual Account > New Token.

Generate and download a registration token.

Install a trust code on the switch:

```
switch# license smart trust idtoken <id_token_value>
```

---

### Step 3: Policy and Reporting

After trust is established, CSSM automatically pushes a policy to the switch.

Optionally adjust reporting interval:

```
switch(config)# license smart usage interval 30
```

---

### Step 4: Verify

Run:

switch# show license status

Confirm transport type = Smart, policy installed, and ACK deadlines populated

Click the information icon to view screenshots and examples.

Click here to view our Smart Licensing Guide

# 3 Connected to SSM On-Prem Through CSSM

Here, switches in the network are connected to SSM On-Prem and SSM On-Prem becomes the single point of interface with CSSM. You can also configure the switch to push the required information to SSM On-Prem.

Switch-initiated communication (push): A switch initiates communication with CSSM by connecting to a REST endpoint in SSM On-Prem. Data that is sent includes RUM reports. Configure the switch to automatically send RUM reports to SSM On-Prem at required intervals.

**Tip for Dell Engineers**

When the SSM On-Prem server is associated with virtual account in CSSM, it will be required that all product instance registration tokens to be generated from their Smart Software On-Prem management interface. After the upgrade, the transport method and the CSLU URL are reset to the default value of cslu-local. Make sure to manually configure the switch with the specific SSM On-Prem URL associated with your Virtual Account (VA). To retrieve the correct URL follow these steps

1. On the On-Prem UI, navigate to Inventory > General.
2. Click CSLU Transport URL.
3. Copy the displayed CSLU Transport URL and use it for configuring the switch.

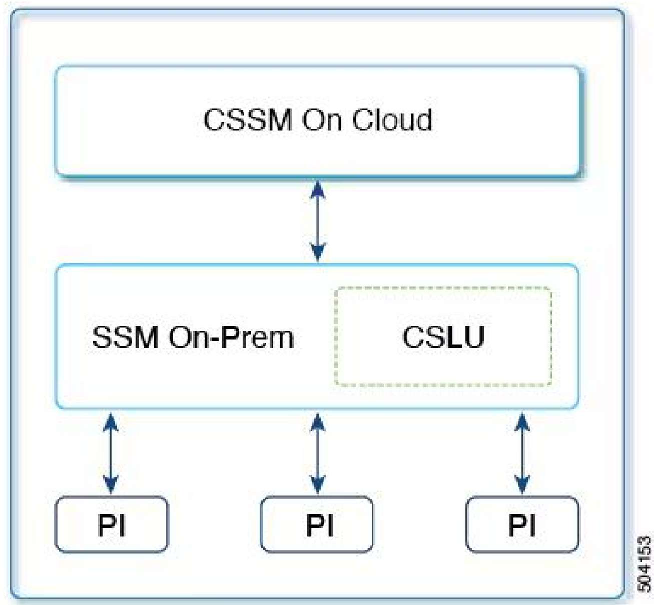


Figure 3 Scenario: Connected to SSM On-Prem Through CSSM

## SLP Configuration - Connected to CSSM Through SSM On-Prem

If the switch is registered On-Prem with pre-SLP release, the transport mode will change to CSLU after the migration. After the upgrade, the transport method and the CSLU URL are reset to the default value of cslu-local. Make sure you manually configure the switch with the specific SSM On-Prem URL associated with your Virtual Account (VA). To retrieve the correct URL:

1. On the On-Prem UI, navigate to Inventory > General.
2. Copy the displayed CSLU Transport URL and use it when configuring the switch.
3. Also, the CSLU URL will be populated on the switch from OnPrem CSLU tenant ID.
4. <tenant_id> is a combination of virtual account and the product instance.
5. Ensure that the configuration is saved using the `copy running-config startup-config` command.

## Step 1: Install SSM On-Prem

1. Download the file from Smart Software Manager > Smart Software Manager On-Prem.
2. Refer to the Cisco Smart License Guide for help with installation and setup.
3. Configure your settings: Log into Cisco (SSM On-Prem Interface)|Configure a Smart Account and a Virtual Account | Add a Product Instance in CSLU

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## Step 2: Configure the Switch

Ensure network reachability from the switch to SSM On-Prem and that transport type is set to cslu. This is the default transport type. If a different option is configured, enter the license smart transport cslu command in global configuration mode. Save any changes to the configuration file.

```
switch(config)# license smart transport cslu
switch(config)# exit
switch# copy running-config startup-config
```

- Specify how SSM On-Prem is to be discovered (choose one):
- Configure a specific URL for SSM On-Prem. If SSM On-Prem was previously configured, then the URL is automatically configured. Otherwise, copy the URL from SSM On-Prem and configure the URL.
- Enter the license smart url cslu http://<ssm_on_prem_ip_or_host>/cslu/v1/pi/<Tenant_ID> , command in global configuration mode. This command can be obtained by the following:
- Log into SSM On-prem web interface.
- Select the correct Account Name.
- Go to Smart Licensing > Inventory
- Under the General tab, click on "CSLU Transport URL" and copy the URL
- For `<ssm_on_prem_ip_or_host>`, enter the hostname or the IP address of the Windows or Linux host where SSM On-Prem is installed.

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## Step 4: Verify

- Run:

Manually configure SSM On-Prem URL:

- Confirm `Cslu address` points to SSM On-Prem URL

Click the information icon to view screenshots and examples.

**Click here to view our Smart Licensing Guide**

# 4 CSLU Disconnected from CSSM

The CSLU utility is installed on-premises and the switches communicate with it. The other side of the communication, between CSLU and CSSM, is offline. In fact, CSLU provides the option of working in a mode that is disconnected from CSSM.
Communication between CSLU and CSSM is sent and received in the form of signed files (tar.gz) that are saved offline and then uploaded to or downloaded from CSLU or CSSM.

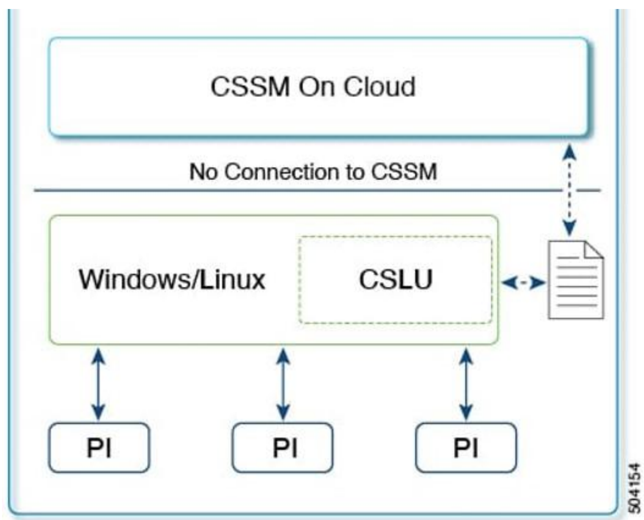


Figure 4 Scenario:  CSLU Disconnected from CSSM

## The key components involved in the process are:

**CSSM On Cloud:** Provides cloud-based software and subscription management.

**CSLU (Cisco Smart Licensing Utility):** Acts as a local proxy for communication with CSSM, especially when direct cloud connectivity is not available.

**Windows/Linux:** The operating system environment where CSLU is installed.

**PI (Product Instance):** Represents a device or software instance that requires licensing.

## The process involves the following stages:

1. PI requests licensing from CSLU.
2. CSLU communicates with CSSM On Cloud to validate and obtain licenses
3. CSSM On Cloud provides license authorization back to CSLU.
4. The saved as tar.gz file in a location that is accessible.
5. CSLU provides license to PI.

Result: CSSM on Cloud enables license management for Product Instances, either directly or through CSLU without connection to CSSM, ensuring software compliance and access to features.

## SLP Configuration - Disconnected from CSSM

### Step 1: Install and Configure CSLU

1. Where the task is performed: download exe, rpm and deb images and deploy it as a VM as per your orchestration environment.
2. Download the file from Smart Software Manager > Smart Licensing Utility.
3. Refer to the Cisco Smart License Utility Quick Start Setup Guide for help with installation and setup.

CSLU Preference Settings | Where tasks are performed: CSLU Interface

- In the CSLU Preferences tab, click the Cisco Connectivity toggle switch to off. The field switches to Cisco Is Not Available.
- Configuring a Smart Account and a Virtual Account
- Adding a Product Instances in CSLU

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### Step 2: Configure the Switch

`Set transport type = CSLU:` `switch(config)# license smart transport cslu`

Configure CSLU discovery:

- Option 1: No action required beyond basic configuration. Name server configured for zero-touch DNS discovery of `cslu-local`.
- Option 2: No action required beyond basic configuration. Name server and domain configured for zero-touch DNS discovery of cslu-local.<domain>.
- Option 3: Configure a specific URL for CSLU. Enter the license smart url cslu `http://<cslu_ip_or_host>:8182/cslu/v1/pi` command in global configuration mode. For <cslu_ip_or_host>, enter the hostname or the IP address of the Windows or Linux host where CSLU is installed. 8182 is the TCP port number and it is the only port number that CSLU uses.

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### Step 3: Export/ Import Licensing Files

1. From CSLU: Data > Export to CSSM → generates RUM file (UD_SA_SA_ACCOUNT_NAME_DATE_TIMESTAMP.tar.gz).
2. In CSSM Web UI: Reports > Usage Data Files > Upload Data → upload RUM file, download ACK.
3. Back in CSLU: Data > Import from CSSM → load ACK file
4. CSLU pushes ACKs to switches automatically.

---

### Step 4: Verify

- On the switch:

```
switch# show license status
```

- Confirm `Last ACK` received shows the acknowledgment file returned from CSSM

**Click here to view our Smart Licensing Guide**

# 5 No Connectivity to CSSM and No CSLU

Here we have a switch and CSSM disconnected from each other without any other intermediary CSLU or components. All communication is in the form of uploaded and downloaded files.

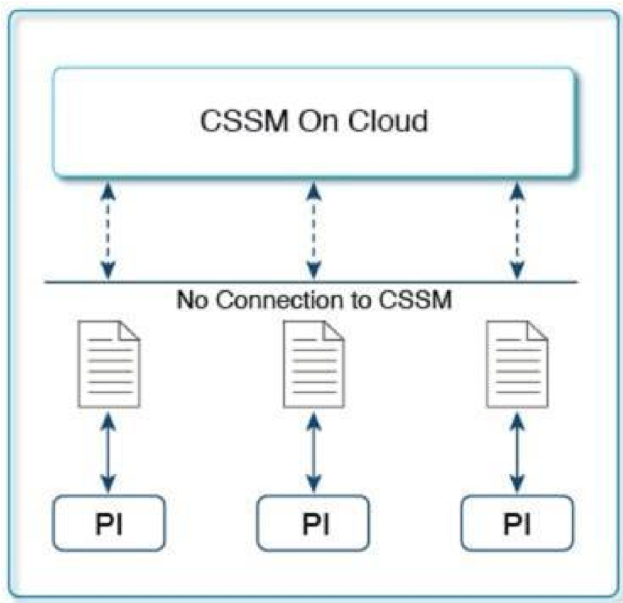


Figure 5. Topology: No Connectivity to CSSM and No CSL

## Configure SLP for No Connectivity to CSSM and No CSLU

This task is applicable when Smart Software Manager (CSSM) connectivity is unavailable and there is no Smart Licensing Utility (CSLU), necessitating local license usage management.

To configure SLP when there is no connectivity to CSSM, perform these steps.

### Step 1: Disable Transport

Use the license smart transport off command in global configuration mode and save any changes to the configuration file.

**Example:**

```
switch(config)# license smart transport off
switch(config)# exit
switch# copy running-config startup-config
```

## Step 2: Generate and Save RUM Reports

Use the license smart save usage command in privileged EXEC mode if any features have been enabled.

Example:

In this example, all RUM reports are saved to the flash memory of the switch, in the all_rum.txt file. The file is first saved to the bootflash and then copied to a TFTP location.

```
switch# license smart save usage all bootflash:all_rum.txt
switch# copy bootflash:all_rum.txt tftp://203.0.113.1/all_rum.txt
```

## Step 3: Upload Usage Data to CSSM

**Uploading Usage Data to CSSM and Downloading an ACK.**

To upload a RUM report to CSSM and download an ACK, perform these steps.

Before you begin | Supported topologies: No Connectivity to CSSM and No CSLU

1. Log in to the CSSM Web UI at https://software.cisco.com/software/smart-licensing/alerts.
2. Log in using the username and password that is provided by Cisco.
3. Select **Reports > Usage Data Files**
4. Click Upload Usage Data. Browse to the file location (RUM report in .txt format), select, and click Upload Data. You cannot delete a usage report in CSSM after it has been uploaded.
5. From the Select Virtual Accounts pop up, select the Virtual Account that will receive the uploaded file. The file is uploaded to Cisco and is listed in the Usage Data Files table in the Reports screen showing the File Name, the time it was Reported, which Virtual Account it was uploaded to, the Reporting Status, the Number of Product Instances reported, and the Acknowledgment status.
6. In the Acknowledgment column, click Download to save the tar.gz acknowledge file for the report that was uploaded. Wait for the Download option to appear in the Acknowledgment column. If multiple RUM reports are being processed, CSSM may take a few minutes to update. You may need to refresh your browser to ensure the latest information is displayed. After the report is available, click Download in the Acknowledgment column to save the acknowledgment (.txt) file for the uploaded report. In the Acknowledgment column, click Download to save the tar.gz acknowledge file for the report that was uploaded. Now, install the file on the switch or transfer it to CSLU or SSM On-Prem.

Click the information icon to view screenshots and examples.

Click here to view our Smart Licensing Guide

## Step 4: Install the Acknowledgement on the Switch

All switch communications are disabled. To allow local RUM-report file saving and uploading to CSSM, you must use a workstation with Internet connectivity to report license usage.

To install a policy or acknowledgment on the switch when the switch is not connected to CSSM, CSLU, or SSM On-Prem, perform these steps.

**Before you begin | Supported topologies | No Connectivity to CSSM and No CSLU**

You must have the corresponding file saved in a location that is accessible to the switch. For a policy, see Downloading a Policy File from CSSM. For an acknowledgment, see Uploading Usage Data to CSSM and Downloading an ACK.

4.1 Copy file from its source location or directory to the flash memory of the switch:
**Example:** `switch#copysourcebootflash:file-name`

| Keyword | Description |
|---|---|
| *source* | This is the location of the source file or directory to be copied. The source can be either local or remote. |
| **bootflash** | This is the destination for boot flash memory. |

4.2 Import and install the file on the switch:

Example:

```
switch# license smart import bootflash:file-name
```

After installation, a system message displays the status of installation.

4.3 Display license authorization, policy, and reporting information for the switch:

**Example:**

```
switch# show license authoization
```

**Click here to view our Smart Licensing Guide**

**These steps provide a comprehensive solution to ensure your customer’s smart accounts deploy corrrectly. Great work!**

# Smart Licensing in Action

The scenarios presented in this guide demonstrate the flexibility of Smart Licensing Using Policy and provide you with a practical roadmap for deployment across a variety of customer environments. By applying these workflows, you will be better equipped to ensure seamless license management, regardless of connectivity or security requirements.

To get a deeper understanding of Cisco Smart Licensing concepts, review the first book in this series, Smart Licensing with Cisco. Together, these resources give you both the conceptual understanding and the practical, step-by-step guidance needed to deliver excellent customer outcomes.

## Here's your path forward:

**Select your Scenario**

- Confirm network restrictions, security requirements, and customer needs.
- Choose the licensing workflow that ensures compliance and minimizes overhead.

**Monitor & Optimize**

- Use Cisco License Central and CSSM to track entitlements, validate usage, and forecast renewals.
- Proactively identify gaps before they impact operations.

**Cisco Software Central**

**Smart Licensing Guide**

**Smart Licensing Hub**

**Licensing Support**

**Smart Licensing with Cisco**

**With Smart Licensing mastered, Dell engineers can deliver clarity, control, and customer trust.**